

MSc in Business Analytics 2025 - 2026



An examination of fraud patterns and predictive modeling

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1. Executive Summary

This report analyses a dataset of 500 customer transactions to identify patterns linked to fraud. Initial data checks showed no missing values and a balanced fraud distribution. Exploratory analysis and statistical tests revealed that most demographic and financial variables such as transaction amount, loan amount, credit score, region, device type and education do not meaningfully differentiate fraud from non-fraud cases. Confidence intervals and hypothesis tests confirmed that these differences are not statistically significant. The only variable showing a significant relationship with fraud is personal income, with fraud victims tending to have lower incomes.

A predictive model (Logistic Regression) achieved moderate performance (ROC-AUC = 0.61), suggesting the dataset lacks strong predictive features. Outlier detection also highlighted unusually large financial losses requiring further review. Recommendations focus on improving behavioral-level monitoring and enhancing data collection.

Overall, the analysis shows that fraud in this dataset is difficult to detect from customer attributes alone and better behavioral-level data is required.

2. Introduction

Fraud detection is an essential part of financial risk management and understanding what factors contribute to fraud can help organisations design better prevention strategies. The analysis follows a structured approach: data cleaning, exploratory data analysis, outlier detection, confidence interval estimation, hypothesis testing and predictive modeling. The goal is to determine whether customer characteristics or transaction behaviors can reliably distinguish fraudulent from non-fraudulent activity.

3. Data Understanding

Understanding the structure and types of variables helps determine which statistical techniques and modeling approaches are appropriate for fraud detection.

3.1 Dataset Overview

The dataset named `fraud_victim_dataset` contains 500 rows and 23 columns. It represents customer transactions where some customers were identified as fraud victims. The target variable is `is_fraud` (1: fraud, 0: non_fraud).

3.2 Variables

Table 1. Variables

Column Name	Data type
<code>transaction_id</code>	object
<code>customer_id</code>	object
<code>age</code>	float64
<code>gender</code>	object
<code>education</code>	object
<code>marital_status</code>	object
<code>region</code>	object
<code>personal_income</code>	float64
<code>family_income</code>	float64
<code>credit_score</code>	int64
<code>loan_amount</code>	float64
<code>has_loan</code>	int64
<code>owns_home</code>	int64
<code>transaction_amount</code>	float64
<code>transaction_type</code>	object
<code>device_type</code>	object
<code>account_age_days</code>	int64
<code>num_prev_transactions</code>	int64
<code>avg_transaction_value</code>	float64
<code>internet_usage_hrs</code>	float64
<code>mobile_usage_hrs</code>	float64
<code>is_fraud</code>	int64
<code>fraud_amount</code>	float64

3.3 Data Quality Check

No missing data were found. No duplicate rows were detected. Most variables are numeric with a few categorical features such as gender, marital_status and region. Since data quality was high, no imputation or removal was required.

3.4 Descriptive Statistics

3.4.1 Numeric Summary

Table 2. Descriptive summary of numeric variables

	count	mean	std	min	25%	50%	75%	max
age	500	50.11	14.42	18.00	39.50	50.20	59.55	90.00
personal_income	500	60214.83	19277.89	15000.00	47652.95	60076.28	72930.48	122754.97
family_income	500	90508.75	23825.70	19922.75	72906.28	90760.21	106926.41	157694.98
credit_score	500	648.61	82.76	411.00	591.00	653.00	707.25	850.00
loan_amount	500	19889.46	20037.90	4.54	5612.26	13875.70	27839.45	115848.45
has_loan	500	0.39	0.49	0.00	0.00	0.00	1.00	1.00
owns_home	500	0.50	0.50	0.00	0.00	0.00	1.00	1.00
transaction_amount	500	2004.38	2095.54	10.00	533.59	1357.21	2846.18	15050.41
account_age_days	500	2586.06	1443.53	32.00	1329.00	2682.00	3786.50	4999.00
num_prev_transactions	500	148.14	87.09	1.00	67.75	145.50	226.00	298.00
avg_transaction_value	500	5119.36	2885.54	75.19	2719.01	5184.03	7633.09	9997.42
internet_usage_hrs	500	4.05	1.95	0.50	2.70	4.00	5.40	10.00
mobile_usage_hrs	500	5.15	2.49	0.50	3.30	5.20	6.80	13.90
is_fraud	500	0.49	0.50	0.00	0.00	0.00	1.00	1.00
fraud_amount	500	712.47	1347.88	0.00	0.00	0.00	841.74	10218.89

Loan amount and transaction amount have very wide ranges. Therefore, investigating data for outliers is necessary. Fraud_amount is highly skewed with most values equal to 0 and a few extremely large ones.

3.4.2 Categorical Summary

Table 3. Descriptive summary of categorical variables

	count	unique	top	freq
transaction_id	500	500	T00001	1
customer_id	500	486	C9889	2
gender	500	2	Female	261
education	500	5	Bachelor	169
marital_status	500	4	Married	230
region	500	4	North	135
transaction_type	500	4	Transfer	131
device_type	500	4	Mobile	137

Each transaction has a unique transaction id. Repeated customer_id values indicated that some customers have multiple transactions. Categorical columns such as gender, education, marital_status, region and device type are well balanced with no dominant category.

4. Data cleaning and preparation

4.1 Handling missing values

There were no missing values in the dataset therefore no data filling was required.

4.2 Detect and Handle outliers

Outliers were detected using the Interquartile Range (IQR), which is widely used for identifying large transactions.

4.2.1 Loan Amounts

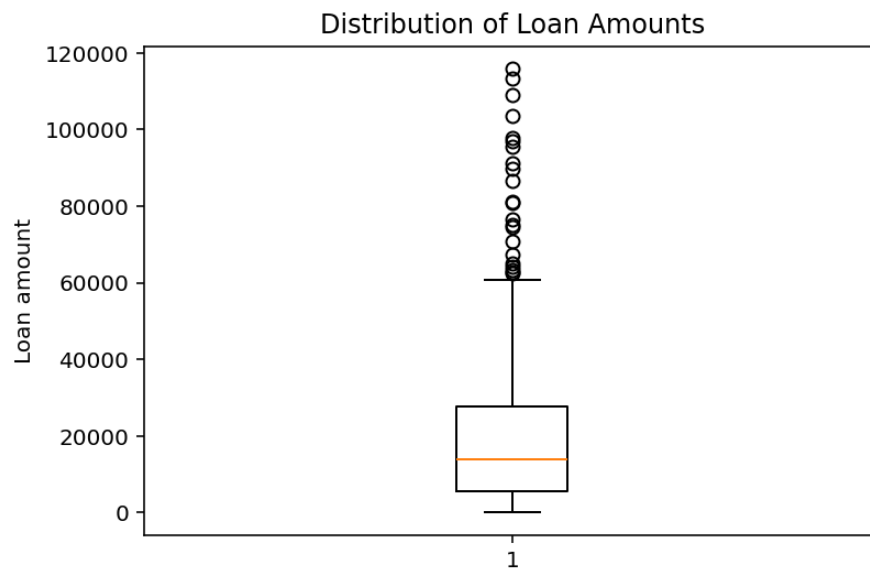


Figure 1. Boxplot showing the distribution of loan amounts.

The boxplot shows several large loan amounts. Using Interquartile Range (IQR), the upper boundary for loan amount was calculated as approximately 61180.

4.2.2 Transaction Amounts

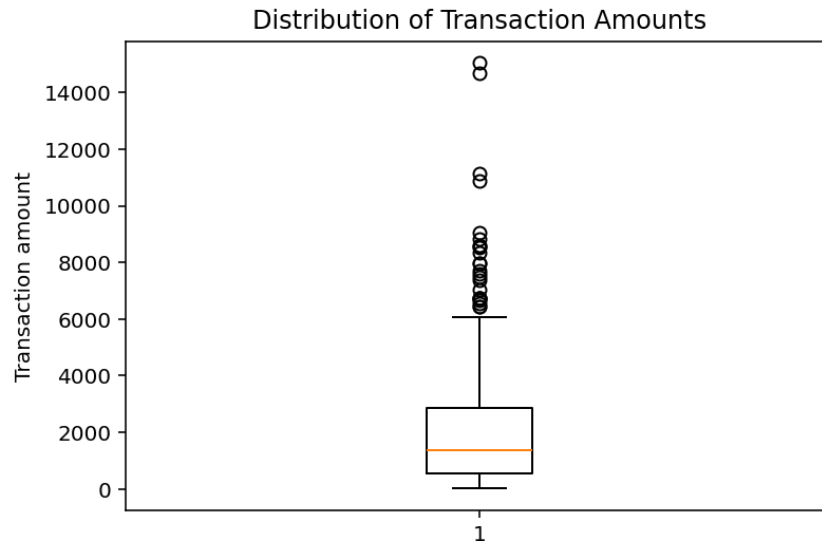


Figure 2. Boxplot showing the distribution of Transaction amounts.

The boxplot shows several large Transaction amounts. Using the Interquartile Range, the upper boundary for transaction amount was calculated as approximately 6315.

4.3 Data Preparation

Numeric features were standardized within the modelling pipeline so that variables on different scales do not dominate the Logistic Regression.

4.4 Summary of Data Cleaning

The data was already clean with no missing or duplicate records. Outliers were found in `loan_amount` and `transaction_amount`. These high values were kept in the data because they might represent fraudulent cases. After checking outliers, the numeric variables were prepared for modeling by standardising them within the pipeline.

5. Exploratory Data Analysis (EDA)

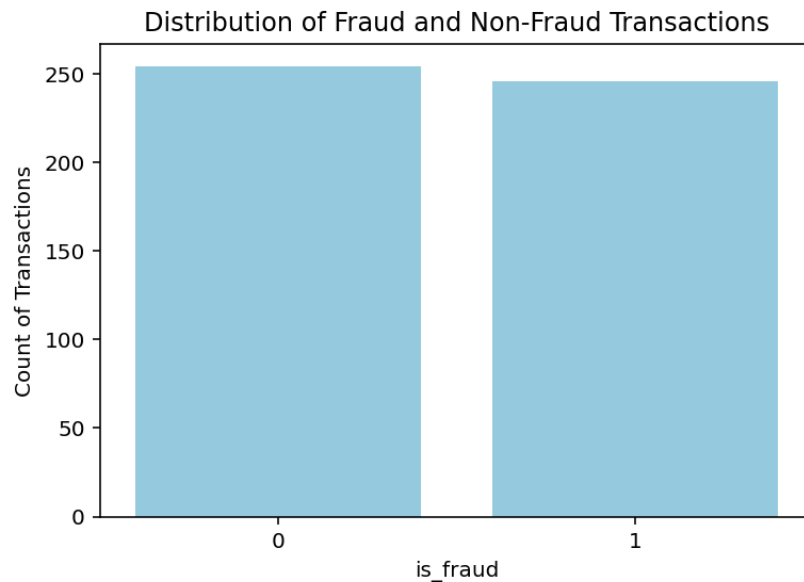


Figure 3. Distribution of Fraud and Non-Fraud Transactions
The bar chart shows that the dataset is almost evenly split between fraudulent and non-fraudulent and the data set is balanced.

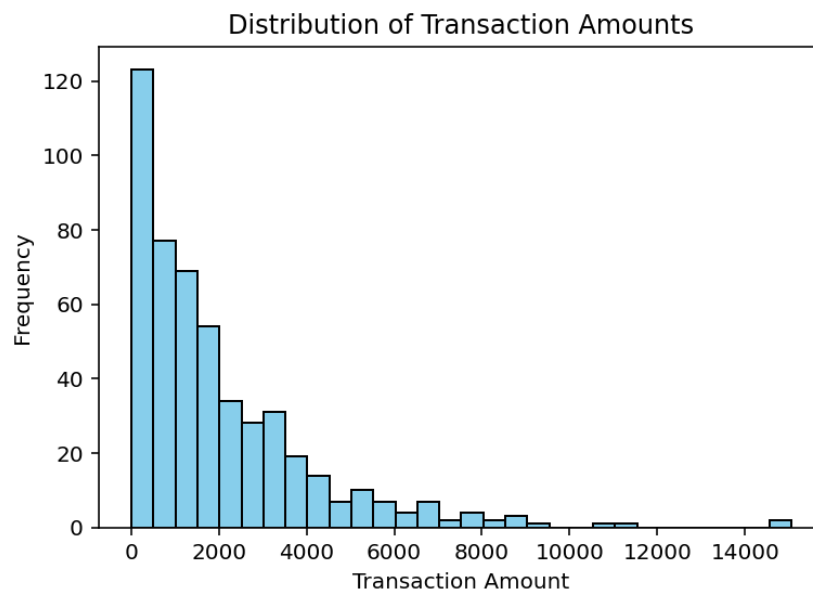


Figure 4. Distribution of Transaction Amount
This histogram shows that most transaction amounts are small with the highest frequency below 2000.

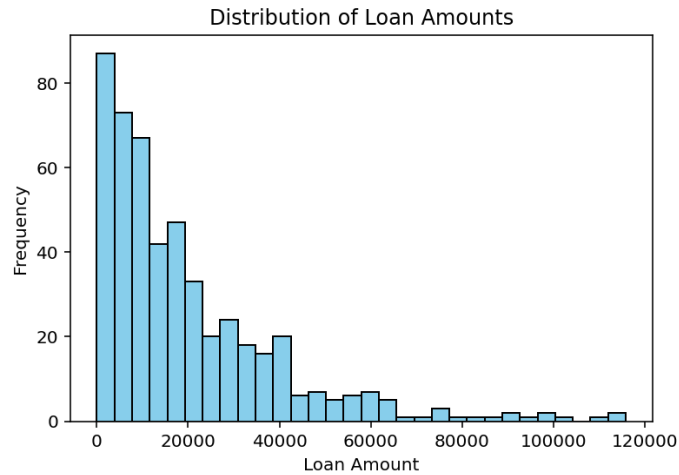


Figure 5. Distribution of Loan Amounts

This histogram shows that most customers have small loan amounts below 20000.

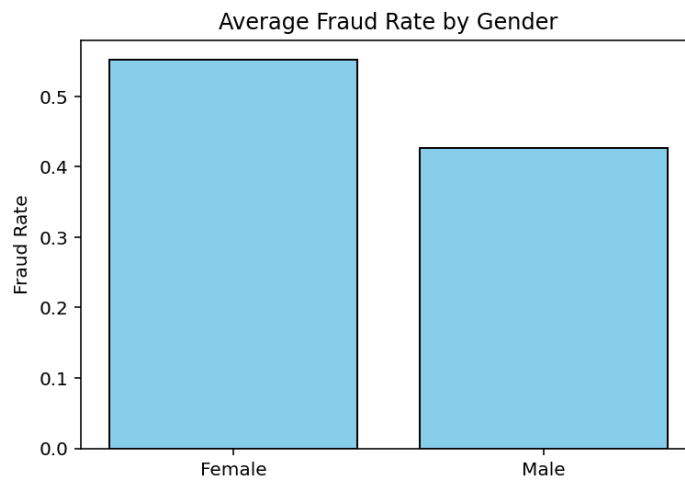


Figure 6. Average Fraud Rate by Gender

The bar chart shows that female customers have slightly higher fraud rate about 55%, compared to male customers (around 43%).

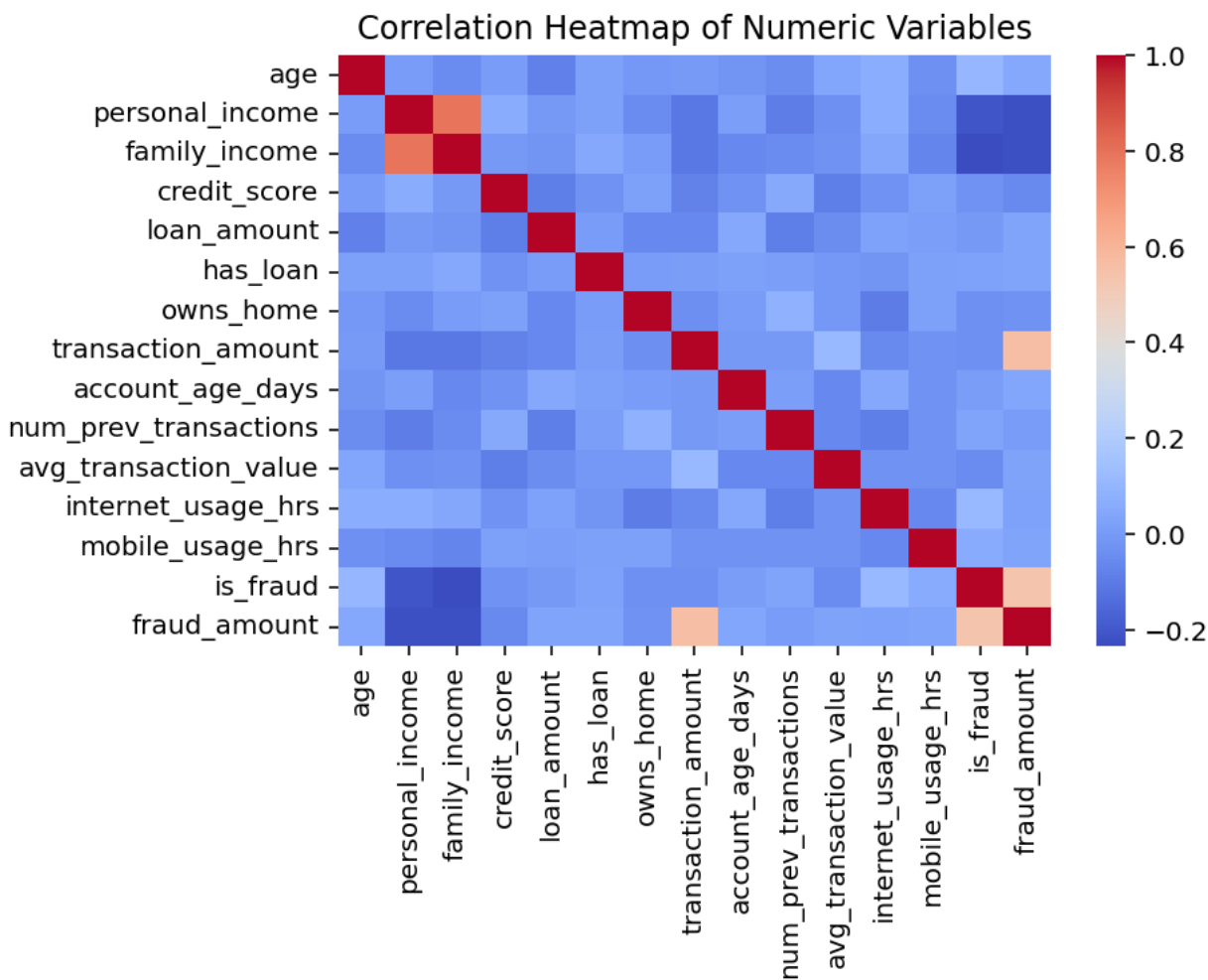


Figure 7. Correlation Heatmap of Numeric Variables

The heatmap shows that there is a strong positive correlation between personal_income and family_income, meaning that people with higher personal income also tend to have higher family income. There is a weak positive relationship between transaction_amount and fraud_amount, suggesting that larger transactions may sometimes be linked to higher fraud amounts. Most other variables show weak or no correlation.

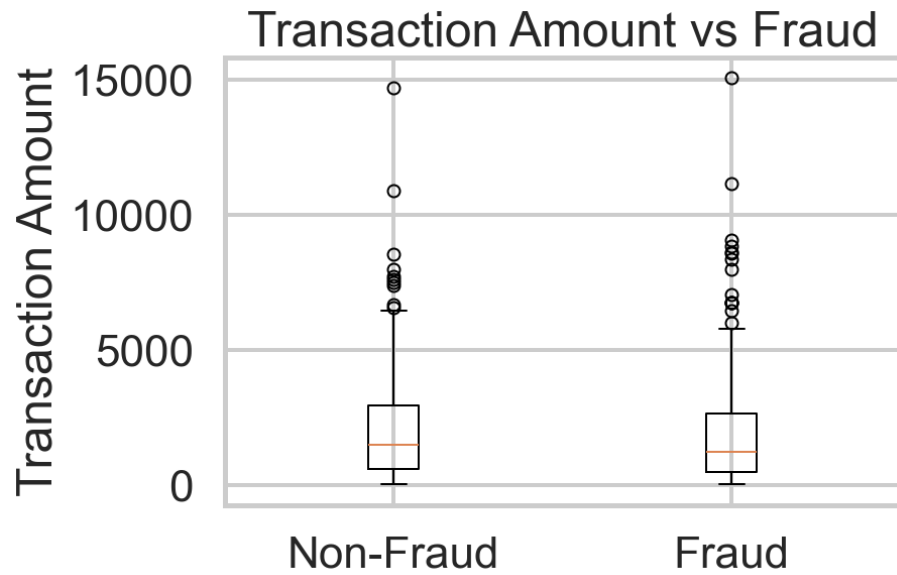


Figure 8. Transaction Amount vs Fraud

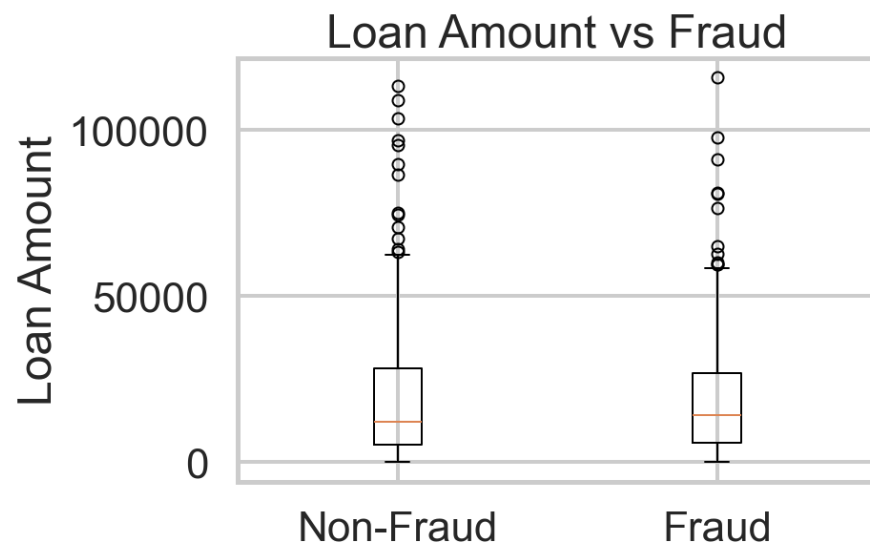


Figure 9. Loan Amount vs Fraud

Both transaction amount and loan amount boxplots show that fraudulent and non-fraudulent customers have very similar medians, spreads and numbers of outliers. This indicates that neither transaction size nor loan size is a strong indicator of fraud in this dataset.

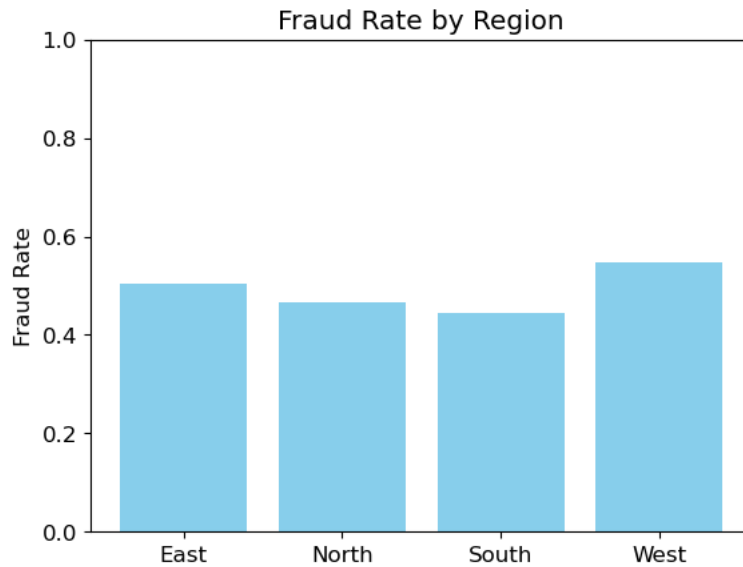


Figure 10. Fraud Rate by region

The fraud rate is fairly similar across all regions, with only a small difference. The west shows the highest fraud rate around 55%, while the south has the lowest around 45%. However, the differences are not large enough to suggest a strong regional pattern.

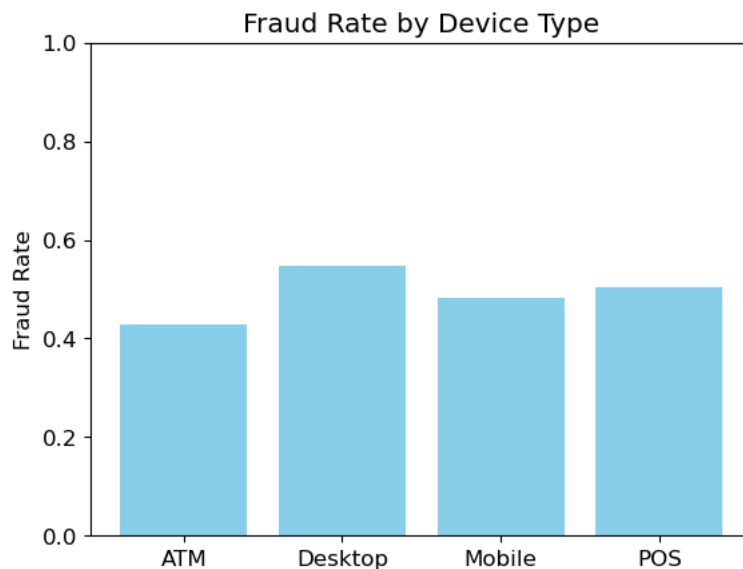


Figure 11. Fraud Rate by Device type

Fraud rates vary slightly across device types, but differences are small. Desktop transactions show the highest fraud rate around 55%, while ATM transactions have the lowest around 43%. Overall, the variation is minor and does not show a strong relationship between the device and likelihood of fraud.

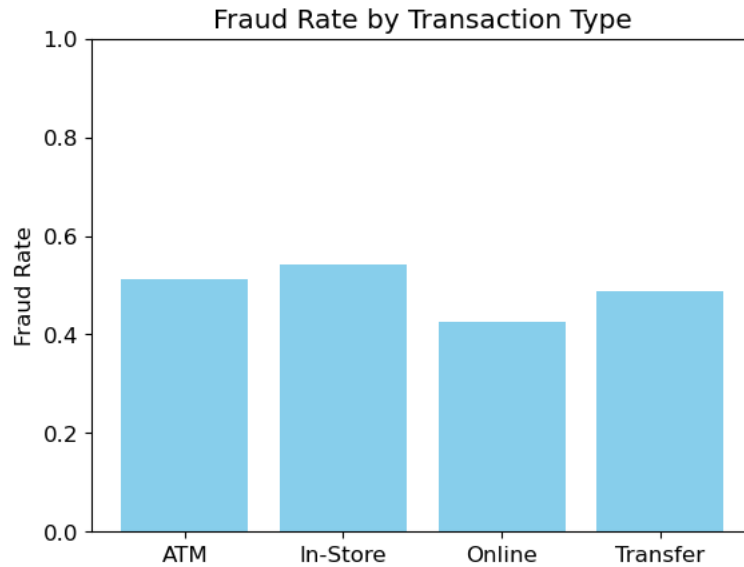


Figure 12. Fraud Rate by Transaction Type

Fraud rates show small differences across transaction types. In-Store has the highest fraud rate around 55%, while Online transactions have the lowest around 50%. Although In-Store transactions appear slightly more risky, the overall differences are minor.



Figure 13. Age vs Fraud

The boxplot shows that fraud and non_fraud customers have very similar age distributions. Both groups have a median age around 50. Fraud cases include slightly more older individuals near the upper range. Overall age does not show a strong pattern.

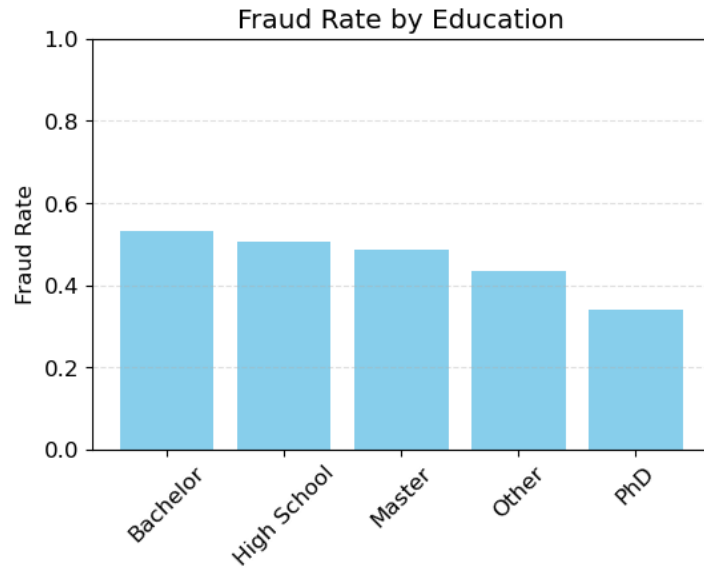


Figure 14. Fraud Rate By Education

Fraud rates vary slightly across education levels, but the differences are relatively small. Bachelor and High School groups have slightly higher fraud rates around 50% to 53%, while customers with PhDs have the lowest rate around 34%. However, the variations are not large enough to suggest the education level is a strong predictor of fraud.

6. Fraud Analysis

6.1. Detecting Outliers

Outliers may represent high-risk or unusual cases. IQR method was used to identify outliers.

6.1.1 Transaction Amount

25 transactions were flagged as potential outliers. These show very large transaction amounts may indicate suspicious cases.

Q1=533.59
Q3=2846.18
IQR = 2312.59
Upper bound=6315.06

6.1.2 Loan Amount

Using IQR method 22 outliers were found. These extreme values may represent high-risk cases.

Q1=5612.26
Q3=27839.45
IQR= 22227.19
Upper Bound: 61180.23

6.1.3 Fraud Amount

The IQR method flags 54 unusually high fraud amounts. These are the biggest losses and should be investigated.

Q1 = 0.0
Q3 = 841.735
IQR = 841.735
Upper Bound = 2104.3375

6.2. Confidence Interval Analysis

A confidence interval gives us a range that likely contains the true average of a population. It helps compare the uncertainty around fraud and non-fraud averages.

6.2.1 CI for mean transaction amount (fraud vs non-fraud)

Fraud and non-fraud transaction sizes fall in almost the same range. Therefore transaction amount alone can not be used to identify fraud and additional features must be analyzed.

Fraud transaction 95% CI : 1661 to 2197

Non-Fraud transaction 95% CI : 1826 to 2329

6.2.2 CI for mean loan amount (fraud vs non-fraud)

The 95% confidence intervals for loan amounts are almost identical. This means the loan amount does not help differentiate fraud from non-fraud.

Fraud 95% CI: 17398.64 to 22151.45

Non-Fraud 95% CI: 17415.6 to 22584.95

6.2.3 CI for fraud rate

The 95% confidence interval shows that the true fraud rate in the population lies between 44.8% and 53.6% , meaning fraud affects roughly half of all transactions.

Fraud rate ($\hat{p}$): 0.492

95% CI for fraud rate: 0.448 to 0.536

6.3 Hypothesis Testing

6.3.1 Personal Income

To examine whether fraud cases are associated with different income levels, a t-test was conducted comparing the mean personal income of fraud and non-fraud customers.

H_0 : There is no difference in mean personal income between fraud and non-fraud customers

H_1 : The mean personal income differs between the two groups

The test result produced a t-statistic of -4.75 and p value of 2.9×10^{-6} , which is far below the 0.05 significance threshold. Therefore we can reject the null hypothesis and conclude the personal income is significantly different across the two groups. Fraudulent Customers tend to have lower personal income.

6.4 Predictive Modeling

Because EDA and statistical tests revealed weak patterns, modelling was used to examine whether a combination of variables can predict fraud more effectively than individual features.

The target variable was `is_fraud` while `fraud_amount`, `transaction_id` and `customer_id` were removed. `Fraud_amount` directly reveals the outcome and should be removed. `Transaction_id` and `customer_id` do not contain predictive information. The model was evaluated using accuracy, precision, recall and ROC-AUC.

Overall, the dataset shows very weak and inconsistent patterns and income is the only factor with significant relationship to fraud. Predictive models achieved moderate accuracy, suggesting that additional behavioral and transactional features would be needed to improve fraud detection.

Accuracy: 0.61

Precision: 0.58

Recall: 0.69

ROC-AUC: 0.61

7. Discussion and Recommendations

7.1 Discussion of Findings

The analysis shows that fraudulent and non-fraudulent customers look very similar across most variables. Transaction amount, loan amount, credit score, account age, education level, region, device type, transaction type all showed weak or no clear patterns. Confidence interval comparisons and hypothesis testing confirmed that none of these differences were statistically significant, meaning these features alone cannot reliably help identify fraud.

The only variable showing a meaningful statistical difference was personal income. Fraud victims tended to have lower personal incomes. While this does not prove low income causes fraud, it suggests that lower income customers may be more vulnerable or more frequently targeted.

Predictive modeling (Logistic Regression) achieved only about 61% accuracy, reflecting that the dataset lacks strong predictors of fraud. No single feature strongly influenced model predictions, and coefficients were small. This further supports that the dataset contains limited information, making fraud hard to predict using the available variables.

Overall, the results show that fraud here is not caused by high amounts or demographics, but likely by hidden factors such as behavior patterns.

7.2 Recommendations

Support Lower-income customers:

Fraud victims tend to have lower personal income. Provide extra prompts for verification checks for this group.

Focus on behavior, not demographics:

Since typical customer attributes do not predict fraud well, use behavioral indicators such as abnormal login activity, device changes, unusual transaction patterns.

Improve data collection:

Current variables offer weak predictive signals. Add richer features such as device information, IP address, log in location, failed login attempts.

Retrain models regularly:

Fraud patterns evolve so models and rules should be updated frequently.